

Goktug Aygun

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EDUCATION

- KU Leuven**, Master of Engineering in Computer Science Sep 2025 – Exp. Jun 2027, Leuven, Belgium
Thesis: When AI Writes Its Own Tests: Evaluating and Improving Test Generation in Coding Agent Loops
- Investigating the reliability of AI-generated tests in coding-agent workflows, focusing on test quality, implementation mirroring, and coverage of developer requirements.
 - Collaborating with Prof. Dr. Tom Holvoet, DistriNet.
- Sabanci University**, B.Sc. in Computer Science and Engineering Sep 2021 – Jun 2025, Istanbul, Turkey
- GPA: 3.78/4.00 | Ranked 26/319 | Sakip Sabanci Encouragement Award 2022 & 2023

EXPERIENCE

- ML6**, AI Engineering Intern Jun 2026 – Sep 2026, Ghent, Belgium
- Architected a multi-agent AI pipeline with a custom file-based API for inter-agent communication, autonomously modernizing over 30,000 lines of financial COBOL across 44 files into modular Python 3
 - Engineered the Python orchestration infrastructure, integrating a deterministic state-machine, Docker-isolated sandbox, and robust pytest-based validation
 - Evaluated execution across 5 repositories using various Claude models, leveraging structured logging to optimize token consumption, cost metrics, and overall migration success rates
- DidactLabs**, Software Engineering Intern Feb 2026 – May 2026, Leuven, Belgium
- Implemented core backend features, resolving critical bugs using TypeScript for an AI-integrated education platform
 - Enforced code quality standards and accelerated development cycles alongside the core engineering team
- OBSS**, Software Engineer Intern Jun 2024 – Sep 2024, Istanbul, Turkey
- Built a responsive React.js and .NET full-stack application, integrating custom REST APIs with strict authentication and authorization
- Sabanci University**, Undergraduate Learning Assistant Sep 2022 – Jan 2024, Istanbul, Turkey
- Mentored 2000+ students in Python and C++ across multiple semesters
 - Led 224 hours of recitation sessions and 336 hours of office hours to reinforce course concepts

PROJECTS

- Retrieval Augmented Generation Search Engine** | *Python* Jan 2026 – Jun 2026
- Integrated retrieval extensions, query rewriting, HyDE, and neural embeddings, to improve context relevance
 - Optimized search accuracy utilizing metadata filtering within a custom RAG architecture
- Multi-Agent Learning in Canonical Games** | *Python, NumPy, PyTorch, Docker* Jan 2026 – Jun 2026
- Simulated multi-agent behaviors in matrix games using Boltzmann Q and epsilon-greedy learning strategies
 - Trained AI agents with integrated computer vision, object detection, and behavior policy management
- Cloud-Deployed Distributed Ticket System** | *Java, Spring Boot, GCP* Sep 2025 – Jan 2026
- Designed and deployed a fault-tolerant, service-oriented ticketing platform on Google Cloud Platform
 - Implemented robust retry, timeout mechanisms ensuring seamless integration with unreliable third-party providers
- LLM-based Course Assistant** | *Python, LLaMA, FAISS, Docker, React* Sep 2024 – Jun 2025
- Fine-tuned LLaMA to deploy a reliable AI course assistant tailored for Sabanci University students and faculty
 - Designed a FAISS-backed RAG pipeline to ensure highly accurate, context-aware educational responses
- Flight Roster Platform** | *Express.js, Node.js, React, AWS, MongoDB* Feb 2024 – Jun 2024
- Developed Node.js backend and RESTful APIs for airline operations platform, leading cross-functional team of 7
 - Deployed scalable MySQL and MongoDB databases on AWS to ensure reliable, high-efficiency data storage

PROFESSIONAL SKILLS

AI & ML: Agentic AI, Langchain, RAG, NLP, Pandas, Matplotlib, Selenium, Hugging Face, PyTorch, Scikit-Learn
Cloud & DevOps: Google Cloud, Docker, Git
Databases: MySQL, MongoDB, NoSQL, Chroma, Firebase
Programming Languages: Python, C++, C#, Java, JavaScript, SQL
Web & Backend Technologies: REST APIs, Flask, Spring, .NET, Node.js, React.js
Languages: Turkish (Native), English (Advanced), Flemish (Beginner), German (Beginner)